ICD - VHDL Interface

XBandDigitalModule_20x20 - ICD VHDL

Document Information

Field	Value
Document ID	XBDM-DOC-009
Version	1.0
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Classification	ITAR Restricted
Status	Preliminary Des

1. Digital Architecture Overview

1.1 Module Hierarchy

```
top_module
+-- jesd204c_receiver
|   +-- lane_sync (x16)
|   +-- frame_align
|   +-- descrambler
+-- ddc_chain
|   +-- nco (x8)
|   +-- cic_decimator (x8)
|   +-- fir_filter (x8)
+-- fifo_buffer
|   +-- async_fifo (x8)
|   +-- sync_fifo
+-- error_handler
|   +-- edac
|   +-- crc
|   +-- fault_log
+-- clock_manager
|   +-- pll_ctrl
|   +-- sysref_gen
+-- spacewire_interface
|   +-- spw_link (x2)
|   +-- spw_codec (x2)
|   +-- spw_router
+-- axi4_lite_slave
    +-- register_file
    +-- irq_controller
```

1.2 Clock Domains

Domain	Frequency	Source	Description
sys_clk	100 MHz	LMK04832-SP	System clock
jesd_clk	10.4 GHz / 66	JESD204C	JESD204C framin
ddc_clk	500 MHz	PLL	DDC processing
spw_clk	100 MHz	SpaceWire	SpaceWire link
axi_clk	100 MHz	LMK04832-SP	AXI bus clock

1.3 Reset Strategy

Reset Source	Type	Duration	Affected Module
power_on	Synchronous	16 cycles	All
software	Synchronous	8 cycles	All except cloc

watchdog	Asynchronous	16 cycles	All
error	Synchronous	8 cycles	error_handler,

2. Register Map

2.1 Register Map Summary

Address Range	Name	Access	Description
0x0000-0x003F	SYS_CTRL	R/W	System control
0x0040-0x007F	ADC_CTRL	R/W	ADC configurati
0x0080-0x00BF	FPGA_CTRL	R/W	FPGA configurat
0x00C0-0x00FF	POWER_CTRL	R/W	Power managemen
0x0100-0x013F	CLOCK_CTRL	R/W	Clock configura
0x0140-0x017F	ERROR_CTRL	R/W	Error handling
0x0180-0x01BF	DIAG_CTRL	R/W	Diagnostics
0x0200-0x02FF	DDC_CTRL	R/W	DDC configurati
0x0300-0x03FF	SPACEWIRE_CTRL	R/W	SpaceWire confi
0x0400-0x04FF	BUFFER_CTRL	R/W	Buffer manageme
0x0500-0x05FF	TRACE_CTRL	R/W	Debug trace
0x1000-0x1FFF	USER_REGS	R/W	User-defined re

2.2 System Control Registers (0x0000-0x003F)

0x0000: SYS_STATUS

Bit	Name	Access	Reset	Description
[31:24]	REVISION	R	0x01	Firmware revisi
[23:16]	STATUS	R	0x00	System status
[15:8]	ERROR	R	0x00	Error flags
[7:0]	VERSION	R	0x00	Version ID

STATUS bits:

ERROR bits:

0x0004: SYS_CTRL

Bit	Name	Access	Reset	Description
[31:28]	RESET	W	0x00	Reset control
[27:24]	ENABLE	R/W	0x00	Module enable
[23:16]	MODE	R/W	0x00	Operating mode
[15:0]	reserved	R/W	0x0000	-

RESET bits (write 1 to assert):

ENABLE bits (write 1 to enable):

MODE bits:

0x0008: SYS_CONFIG

Bit	Name	Access	Reset	Description
[31:24]	reserved	R/W	0x00	-
[23:16]	WATCHDOG	R/W	0xFF	Watchdog timeou
[15:8]	TEMPOUT	R/W	0x80	Temperature ala
[7:0]	DEBUG	R/W	0x00	Debug mode

0x000C: SYS_IRQ_STATUS

Bit	Name	Access	Reset	Description
[31:16]	reserved	R	0x0000	-
[15]	IRQ_TEMP	R	0	Temperature ala
[14]	IRQ_OVERFLOW	R	0	Data overflow
[13]	IRQ_ERROR	R	0	System error
[12]	IRQ_JESD	R	0	JESD204C status
[11]	IRQ_SPW	R	0	SpaceWire event
[10]	IRQ_WDOG	R	0	Watchdog timeou

[9:0]	reserved	R	0x000	-

0x0010: SYS_IRQ_ENABLE

Bit	Name	Access	Reset	Description
[31:16]	reserved	R/W	0x0000	-
[15]	EN_IRQ_TEMP	R/W	0	Enable temperat
[14]	EN_IRQ_OVERFLOW	R/W	0	Enable overflow
[13]	EN_IRQ_ERROR	R/W	0	Enable error IR
[12]	EN_IRQ_JESD	R/W	0	Enable JESD IRQ
[11]	EN_IRQ_SPW	R/W	0	Enable SpaceWir
[10]	EN_IRQ_WDOG	R/W	0	Enable watchdog
[9:0]	reserved	R/W	0x000	-

2.3 ADC Control Registers (0x0040-0x007F)

0x0040: ADC_CTRL

Bit	Name	Access	Reset	Description
[31:28]	DECIMATION	R/W	0x0	Decimation fact
[27:24]	GAIN	R/W	0x0	Digital gain (0
[23:20]	TEST_MODE	R/W	0x0	Test pattern (0
[19:16]	FORMAT	R/W	0x0	Data format (0:
[15:12]	reserved	R/W	0x0	-
[11:8]	JESD_MODE	R/W	0x0	JESD mode (0:20
[7:4]	LANE_EN	R/W	0xF	Lane enable (bi
[3:0]	reserved	R/W	0x0	-

0x0044: ADC_STATUS

Bit	Name	Access	Reset	Description
[31:24]	JESD_STATUS	R	0x00	JESD204C status
[23:16]	FIFO_LEVEL	R	0x00	FIFO fill level
[15:8]	ERROR	R	0x00	ADC error flags
[7:0]	reserved	R	0x00	-

JESD_STATUS bits:

ERROR bits:

0x0048: ADC_NCO_FREQ

Bit	Name	Access	Reset	Description
[31:0]	FREQ_WORD	R/W	0x00000000	NCO frequency w

Calculation: FREQ_WORD = (f_IF / f_sample) x 2^{32}

0x004C: ADC_NCO_PHASE

Bit	Name	Access	Reset	Description
[31:16]	PHASE_OFF	R/W	0x0000	NCO phase offse
[15:0]	reserved	R/W	0x0000	-

2.4 FPGA Control Registers (0x0080-0x00BF)

0x0080: FPGA_TEMP

Bit	Name	Access	Reset	Description
[31:16]	reserved	R	0x0000	-
[15:8]	TEMP_RAW	R	0x00	Raw temperature
[7:0]	TEMP Converted	R	0x00	Temperature (d

0x0084: FPGA_VOLTAGE

Bit	Name	Access	Reset	Description
[31:16]	reserved	R	0x0000	-
[15:8]	VCCINT	R	0x00	Core voltage (8
[7:0]	VCCAUX	R	0x00	Aux voltage (8-

0x0088: FPGA_CONFIG

Bit	Name	Access	Reset	Description
[31:24]	VERSION	R	0x00	FPGA version
[23:16]	BUILD_DATE	R	0x00	Build date
[15:8]	BUILD_TIME	R	0x00	Build time
[7:0]	CONFIG_DONE	R	0x00	Configuration s

2.5 Power Control Registers (0x00C0-0x00FF)

0x00C0: POWER_STATUS

Bit	Name	Access	Reset	Description
[31:28]	RAIL_STATUS	R	0xF	Rail power-good
[27:24]	CURRENT	R	0x0	Current (4-bit,
[23:16]	reserved	R	0x00	-
[15:8]	SEQUENCER_STATU	R	0x00	Sequencer statu
[7:0]	FAULT_STATUS	R	0x00	Fault status

RAIL_STATUS bits:

0x00C4: POWER_CTRL

Bit	Name	Access	Reset	Description
[31:28]	RAIL_ENABLE	R/W	0x0	Rail enable (bi
[27:24]	SEQUENCER_CTRL	R/W	0x0	Sequencer contr
[23:16]	CURRENT_LIMIT	R/W	0x0	Current limit s
[15:0]	reserved	R/W	0x0000	-

2.6 Clock Control Registers (0x0100-0x013F)

0x0100: CLOCK_STATUS

Bit	Name	Access	Reset	Description
[31:28]	PLL_LOCKED	R	0x0	PLL lock status
[27:24]	reserved	R	0x0	-
[23:16]	JITTER	R	0x00	Jitter (8-bit,
[15:0]	FREQ_ACTUAL	R	0x0000	Actual frequenc

0x0104: CLOCK_CTRL

Bit	Name	Access	Reset	Description
[31:28]	PLL_RESET	W	0x0	PLL reset (writ
[27:24]	PLL_ENABLE	R/W	0x0	PLL enable
[23:16]	DIVIDER	R/W	0x00	Output divider
[15:0]	reserved	R/W	0x0000	-

0x0108: CLOCK_NCO_FREQ

Bit	Name	Access	Reset	Description
[31:0]	FREQ_WORD	R/W	0x00000000	Clock frequency

2.7 Error Control Registers (0x0140-0x017F)

0x0140: ERROR_STATUS

Bit	Name	Access	Reset	Description
[31:24]	ADC_ERROR	R	0x00	ADC error count
[23:16]	FPGA_ERROR	R	0x00	FPGA error coun
[15:8]	CLOCK_ERROR	R	0x00	Clock error cou
[7:0]	POWER_ERROR	R	0x00	Power error cou

0x0144: ERROR_CTRL

Bit	Name	Access	Reset	Description
[31:28]	ERROR_MASK	R/W	0xF	Error mask (bit
[27:24]	ERROR_ACTION	R/W	0x0	Error action (0
[23:16]	reserved	R/W	0x00	-
[15:0]	ERROR_THRESHOLD	R/W	0x0100	Error threshold

0x0148: ERROR_LOG

Bit	Name	Access	Reset	Description
[31:16]	TIMESTAMP	R	0x0000	Error timestamp
[15:8]	ERROR_TYPE	R	0x00	Error type
[7:0]	ERROR_SOURCE	R	0x00	Error source

2.8 Diagnostics Control Registers (0x0180-0x01BF)

0x0180: DIAG_CTRL

Bit	Name	Access	Reset	Description
[31:28]	TEST_PATTERN	R/W	0x0	Test pattern se
[27:24]	LOOPBACK	R/W	0x0	Loopback mode
[23:16]	PRBS_EN	R/W	0x0	PRBS generator
[15:0]	reserved	R/W	0x0000	-

0x0184: DIAG_STATUS

Bit	Name	Access	Reset	Description
[31:24]	PATTERN_ERR	R	0x00	Pattern error c
[23:16]	BIT_ERR	R	0x00	Bit error count
[15:8]	FRAME_ERR	R	0x00	Frame error cou
[7:0]	reserved	R	0x00	-

2.9 DDC Control Registers (0x0200-0x02FF)

8 DDC channels, each with 16 registers (0x20 bytes per channel)

Channel	Base Address	Registers
DDC0	0x0200	0x0200-0x021F
DDC1	0x0220	0x0220-0x023F
DDC2	0x0240	0x0240-0x025F
DDC3	0x0260	0x0260-0x027F
DDC4	0x0280	0x0280-0x029F
DDC5	0x02A0	0x02A0-0x02BF
DDC6	0x02C0	0x02C0-0x02DF
DDC7	0x02E0	0x02E0-0x02FF

DDC Channel Register Map (per channel)

Offset	Name	Access	Description
0x00	DDC_CTRL	R/W	Channel control
0x04	DDC_STATUS	R	Channel status
0x08	DDC_FREQ_LOW	R/W	Frequency word
0x0C	DDC_FREQ_HIGH	R/W	Frequency word
0x10	DDC_PHASE	R/W	Phase offset
0x14	DDC_GAIN	R/W	Gain control
0x18	DDC_OFFSET_I	R/W	I offset
0x1C	DDC_OFFSET_Q	R/W	Q offset

DDC_CTRL bits:

DDC_STATUS bits:

2.10 SpaceWire Control Registers (0x0300-0x03FF)

0x0300: SPW_STATUS

Bit	Name	Access	Reset	Description
[31:28]	LINK_STATUS	R	0x0	Link status (bi
[27:24]	reserved	R	0x0	-
[23:16]	TX_COUNT	R	0x00	TX packet count
[15:8]	RX_COUNT	R	0x00	RX packet count
[7:0]	ERROR_COUNT	R	0x00	Error count

LINK_STATUS bits:

0x0304: SPW_CTRL

Bit	Name	Access	Reset	Description
[31:28]	LINK_ENABLE	R/W	0x0	Link enable
[27:24]	LINK_RESET	W	0x0	Link reset
[23:16]	TX_ENABLE	R/W	0x0	TX enable
[15:8]	RX_ENABLE	R/W	0x0	RX enable
[7:0]	reserved	R/W	0x00	-

0x0308: SPW_CONFIG

Bit	Name	Access	Reset	Description
[31:24]	TX_THRESHOLD	R/W	0x08	TX threshold
[23:16]	RX_THRESHOLD	R/W	0x08	RX threshold
[15:8]	TIMEOUT	R/W	0x64	Timeout (ms)
[7:0]	reserved	R/W	0x00	-

0x030C-0x03FF: SPW_ROUTER

Offset	Name	Access	Description
0x030C	ROUTER_CONFIG	R/W	Router configur
0x0310	ROUTER_TABLE	R/W	Routing table (
0x03F0	ROUTER_STATUS	R	Router status

2.11 Buffer Control Registers (0x0400-0x04FF)

0x0400: BUFFER_STATUS

Bit	Name	Access	Reset	Description
[31:24]	FIFO_LEVEL	R	0x00	FIFO fill level
[23:16]	FIFO_FREE	R	0xFF	FIFO free space
[15:8]	OVERFLOW_COUNT	R	0x00	Overflow count
[7:0]	UNDERFLOW_COUNT	R	0x00	Underflow count

0x0404: BUFFER_CTRL

Bit	Name	Access	Reset	Description
[31:28]	FIFO_RESET	W	0x0	FIFO reset
[27:24]	FIFO_ENABLE	R/W	0x0	FIFO enable
[23:16]	THRESHOLD_HIGH	R/W	0xC0	High threshold
[15:8]	THRESHOLD_LOW	R/W	0x40	Low threshold
[7:0]	reserved	R/W	0x00	-

2.12 Trace Control Registers (0x0500-0x05FF)

0x0500: TRACE_CTRL

Bit	Name	Access	Reset	Description
[31:28]	TRACE_ENABLE	R/W	0x0	Trace enable
[27:24]	TRACE_SOURCE	R/W	0x0	Trace source se
[23:16]	TRACE_TRIGGER	R/W	0x00	Trace trigger
[15:0]	TRACE_SIZE	R/W	0x0100	Trace buffer si

0x0504: TRACE_STATUS

Bit	Name	Access	Reset	Description
[31:16]	TRACE_COUNT	R	0x0000	Current trace c
[15:8]	TRACE_STATE	R	0x00	Trace state
[7:0]	reserved	R	0x00	-

3. Interrupt Architecture

3.1 Interrupt Sources

Source	Priority	Vector	Description
Temperature ala	High	0x01	Overtemperature
Power fault	High	0x02	Power supply er
Clock unlock	High	0x03	PLL lost lock

JESD error | Medium | 0x10 | JESD204C error
SpaceWire event | Medium | 0x20 | SpaceWire link
Data overflow | Medium | 0x30 | FIFO overflow
Watchdog | Low | 0x40 | Watchdog timeou
Debug | Low | 0x50 | Debug event

3.2 Interrupt Flow

```
Interrupt Source --? Priority Encoder --? IRQ Controller --? ARM Cortex-R5F
                                          |
                                          +--? IRQ_STATUS register
                                          +--? IRQ_ENABLE mask
                                          +--? IRQ_CLEAR (write 1 to clear)
```

4. VHDL Module Interface Summary

4.1 top_module

```
entity top_module is
    port (
        -- Clock and Reset
        sys_clk         : in  std_logic;                      -- 100 MHz system clock
        sys_rst_n       : in  std_logic;                      -- Active low reset

        -- JESD204C Interface (to ADC)
        jesd_refclk_p   : in  std_logic;                      -- JESD reference clock
        jesd_refclk_n   : in  std_logic;
        jesd_tx_p       : in  std_logic_vector(15 downto 0);  -- JESD TX data
        jesd_tx_n       : in  std_logic_vector(15 downto 0);
        jesd_rx_p       : out std_logic_vector(15 downto 0);  -- JESD RX data
        jesd_rx_n       : out std_logic_vector(15 downto 0);
        jesd_sync_n     : out std_logic;                      -- JESD sync
        jesd_sysref     : in  std_logic;                      -- JESD SYSREF

        -- SpaceWire Interface
        spw0_dout_p     : out std_logic;                      -- SpaceWire 0 data+
        spw0_dout_n     : out std_logic;                      -- SpaceWire 0 data-
        spw0_sout_p     : out std_logic;                      -- SpaceWire 0 strobe+
        spw0_sout_n     : out std_logic;                      -- SpaceWire 0 strobe-
        spw0_din_p      : in  std_logic;                      -- SpaceWire 0 data+
        spw0_din_n      : in  std_logic;                      -- SpaceWire 0 data-
        spw0_sin_p      : in  std_logic;                      -- SpaceWire 0 strobe+
        spw0_sin_n      : in  std_logic;                      -- SpaceWire 0 strobe-
        spw1_dout_p     : out std_logic;                      -- SpaceWire 1 data+
        spw1_dout_n     : out std_logic;                      -- SpaceWire 1 data-
        spw1_sout_p     : out std_logic;                      -- SpaceWire 1 strobe+
        spw1_sout_n     : out std_logic;                      -- SpaceWire 1 strobe-
        spw1_din_p      : in  std_logic;                      -- SpaceWire 1 data+
        spw1_din_n      : in  std_logic;                      -- SpaceWire 1 data-
        spw1_sin_p      : in  std_logic;                      -- SpaceWire 1 strobe+
        spw1_sin_n      : in  std_logic;                      -- SpaceWire 1 strobe-

        -- SPI Interface (to Clock/Power)
        spi_mosi        : out std_logic;                      -- SPI master out
        spi_miso        : in  std_logic;                      -- SPI master in
        spi_sclk        : out std_logic;                      -- SPI clock
        spi_cs_n        : out std_logic_vector(3 downto 0);   -- SPI chip select

        -- I2C Interface (to Sensors)
        i2c_scl         : out std_logic;                      -- I2C clock
        i2c_sda         : inout std_logic;                    -- I2C data

        -- GPIO
        gpio            : inout std_logic_vector(7 downto 0); -- General purpose I/O

        -- Debug
        uart_tx         : out std_logic;                      -- UART TX
```

```
        uart_rx         : in  std_logic;                      -- UART RX
```

4.2 jesd204c_receiver

```
entity jesd204c_receiver is
    generic (
        NUM_LANES   : integer := 16;
        DATA_WIDTH  : integer := 32
    );
    port (
        clk         : in  std_logic;
        rst_n       : in  std_logic;
        -- JESD204C physical
        rx_p        : in  std_logic_vector(NUM_LANES-1 downto 0);
        rx_n        : in  std_logic_vector(NUM_LANES-1 downto 0);
        refclk_p    : in  std_logic;
        refclk_n    : in  std_logic;
        sync_n      : out std_logic;
        sysref      : in  std_logic;
        -- Parallel data output
        data_out    : out std_logic_vector(DATA_WIDTH-1 downto 0);
        data_valid  : out std_logic;
        -- Status
        status      : out std_logic_vector(31 downto 0)
    );
end entity jesd204c_receiver;
```

4.3 ddc_chain

```
entity ddc_chain is
    generic (
        NUM_CHANNELS : integer := 8;
        DATA_WIDTH   : integer := 16;
        NCO_WIDTH    : integer := 48
    );
    port (
        clk         : in  std_logic;
        rst_n       : in  std_logic;
        -- Input
        data_in     : in  std_logic_vector(DATA_WIDTH-1 downto 0);
        data_valid  : in  std_logic;
        -- Output (I/Q per channel)
        i_out       : out std_logic_vector(NUM_CHANNELS*DATA_WIDTH-1 downto 0);
        q_out       : out std_logic_vector(NUM_CHANNELS*DATA_WIDTH-1 downto 0);
        out_valid   : out std_logic;
        -- Control
        nco_freq    : in  std_logic_vector(NUM_CHANNELS*NCO_WIDTH-1 downto 0);
        nco_phase   : in  std_logic_vector(NUM_CHANNELS*16-1 downto 0);
        gain        : in  std_logic_vector(NUM_CHANNELS*4-1 downto 0);
        decimation  : in  std_logic_vector(NUM_CHANNELS*4-1 downto 0)
    );
end entity ddc_chain;
```

5. Revision History

Version	Date	Author	Description
1.0	2026-09-09	FPGA/VHDL Engin	Initial release
